This Robot Bird Can Fly -- and Swim Underwater

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Diving birds, such as Atlantic puffins, do not use their wings just for flying. They also use them to swim underwater and chase **prey**. Inspired by these birds, a team of scientists led by Raphael Zufferey of MIT has created a robot that does the same thing.

The lightweight robot is called the FAAV, short for flapping-wing aerial-aquatic vehicle. It uses its wings to move smoothly between flying and swimming. Building a machine that can move through both air and water was no easy task. Water is about 1,000 times **denser** than air, so moving through it requires much more force. Birds can fold their wings to help them swim, but adding a folding **mechanism** would make a small robot more complicated. To solve this, the team gave the robot **flexible** wings that bend up to 90 degrees when pushing through water.

Next, the scientists had to find the best wing size. They tested three different sizes and found that medium-sized wings offered a good balance. They bent enough to work underwater while still providing the lift needed for flight. To help the wings shed water after swimming, they were coated with tiny, **water-repelling** particles. This kept them light enough for takeoff.

In tests, the robot swam underwater at nearly 3 feet (1 meter) per second and flew at about 20 feet (6 meters) per second. These speeds were similar to those of real diving birds. Also, unlike most diving birds, the robot did not need to paddle its feet to get airborne. It rose out of the water and took off using only its wings.

The team published its findings in the journal Science on July 9, 2026. The researchers are now working to improve the wings so they can turn as well as flap up and down. They also plan to test the robot in choppy water and windy conditions. Once ready, the robot could help scientists study the ocean more efficiently.

"One of the major challenges in ocean science is collecting data both frequently and across many locations, which is something this robot could do in the future," Zufferey says. "You could send this out not just every week, but every hour. It could fly out at high speeds, dive in, fly back, deliver its data, and go back out, multiple times."

Word Treasure

prey -- An animal that is hunted and eaten by another animal.
denser -- Packed more tightly together, so it is harder to push through.
mechanism -- A set of moving parts that work together to do a job.
flexible -- Able to bend easily without breaking.
water-repelling -- Pushing water away so it does not soak in or stick.

Background you need

Some seabirds, like puffins, are built for two worlds: they fly through the air and also 'fly' underwater, using their wings to chase fish.

Engineers often copy nature when they design machines, because animals have already solved hard problems like moving smoothly or saving energy.

A machine that can both fly and swim could be useful for ocean science, since much of the sea is hard to reach and study from a boat or from shore.

The hard part is that air and water are very different: water pushes back far more strongly, so a wing that works well in one may fail in the other.

Break it down

LEAD: Scientists at MIT built a robot bird that flies through the air and swims underwater using the same flapping wings.

KEY EVENT: The team created the FAAV, a lightweight robot inspired by diving birds like Atlantic puffins.

+- WHO: A team led by Raphael Zufferey of MIT

+- CHALLENGE: Water is about 1,000 times denser than air, so moving through it takes much more force.

+- SOLUTION: Flexible wings that bend up to 90 degrees; medium-sized wings worked best; a water-repelling coating keeps them light.

+- RESULTS: It swam at nearly 3 feet per second and flew at about 20 feet per second, like real diving birds, and took off without paddling.

EVIDENCE: The team published its findings in the journal Science on July 9, 2026.

OPEN QUESTION: Can the wings be improved to turn as well as flap, and can the robot handle choppy water and wind?

Why it matters

A robot that can fly and swim could help scientists gather ocean data far more often and from far more places, which matters because the ocean is huge, hard to reach, and key to understanding our planet.

Test what you remember

1. What inspired the robot's design?

- (A) Fish that jump out of the water
- (B) Airplanes with folding wings
- (C) Diving birds such as Atlantic puffins
- (D) Bats that hunt at night

2. Why is moving through water harder than moving through air?

- (A) Water has more salt than air
- (B) Water is colder than air
- (C) Water is darker than air
- (D) Water is about 1,000 times denser than air

3. How do the robot's wings handle swimming?

- (A) They come off after takeoff
- (B) They spin like a propeller
- (C) They bend up to 90 degrees
- (D) They fold flat against its body

4. Which wing size worked best in the tests?

- (A) Medium-sized wings
- (B) Wings of all three sizes worked equally well
- (C) The largest wings
- (D) The smallest wings

5. How fast did the robot swim and fly in tests?

- (A) 3 feet per second in the air and 20 feet per second underwater
- (B) About 1 foot per second in both air and water
- (C) Nearly 3 feet per second underwater and about 20 feet per second in the air
- (D) Nearly 90 feet per second in the air

6. What makes the robot different from most diving birds when taking off?

- (A) It needs a running start on the water
- (B) It takes off using only its wings, without paddling its feet
- (C) It flaps its wings and paddles its feet at the same time
- (D) It takes off from dry land only

Answer key (try the quiz first!): 1: C 2: D 3: C 4: A 5: C 6: B

Questions to think about

1. Raphael Zufferey: Says the robot could one day collect ocean data every hour, flying out, diving in, and returning again and again.

MY THOUGHTS (at least 20 words):